CORRECTION NEUROPHYSIOLOGY MIDTERM, PRACTICE 2

Correct answers in red, with a short explanation.

1- A patient receives a drug that stimulates muscarinic receptors. Which effect is expected?

A- Reduced gut motility
B- Dilation of the bronchi
C- Faster heart rate
D- Dry mouth
E- Constriction of the pupils

Answer E. Muscarinic receptors mediate parasympathetic effects: miosis, copious saliva, bronchoconstriction, slower heart, more gut motility. ***Slide:*** *Intro to neurophysiology, slide 31.*

2- The adrenal medulla differs from other sympathetic targets because it

A- Is supplied directly by preganglionic fibres
B- Receives postganglionic cholinergic fibres
C- Releases noradrenaline and adrenaline in a 1:4 ratio
D- Is supplied by parasympathetic fibres
E- Releases mainly dopamine

Answer A. The adrenal medulla acts like a sympathetic ganglion and releases adrenaline and noradrenaline about 4:1. ***Slide:*** *Intro to neurophysiology, slide 40.*

3- At all autonomic ganglia, the neurotransmitter and receptor are

A- Dopamine on D1 receptors
B- Acetylcholine on nicotinic receptors
C- Acetylcholine on muscarinic receptors
D- Noradrenaline on beta receptors
E- Noradrenaline on alpha receptors

Answer B. ACh acts on nicotinic receptors at all autonomic ganglia and on muscarinic receptors at parasympathetic terminals. ***Slide:*** *Intro to neurophysiology, slide 36.*

4- A child has a tumour that blocks the cerebral aqueduct. Which ventricles are expected to dilate?

A- The fourth ventricle only
B- The lateral ventricles only
C- All four ventricles
D- The lateral ventricles and the third ventricle
E- The third and fourth ventricles

Answer D. CSF made upstream of the aqueduct cannot leave, so the lateral and third ventricles dilate; the fourth is downstream. ***Slide:*** *Intro to neurophysiology, slides 28 and 30.*

5- A blind student reads Braille with her fingertips. Detecting the shape and texture of the dots relies mostly on

A- Merkel discs
B- Pacinian corpuscles
C- Ruffini endings
D- Muscle spindles
E- Nociceptors

Answer A. Merkel discs are slowly adapting and detect form and texture; they make up about 25% of hand receptors. ***Slide:*** *Intro to neurophysiology, slide 51.*

6- Rubbing a bumped elbow relieves the pain. According to the gate control theory, this is because

A- Nociceptors adapt within milliseconds
B- A-beta touch fibres activate inhibitory interneurons in the dorsal horn
C- The motor cortex blocks the thalamus
D- The pain fibres are cut
E- C fibres release more substance P

Answer B. Large touch fibres (A-beta) and descending enkephalin neurons close the 'gate' on pain transmission in the dorsal horn. ***Slide:*** *Intro to neurophysiology, slide 62.*

7- The nicotinic acetylcholine receptor of the motor end plate is an example of a

A- Voltage-gated channel
B- Mechanically gated channel
C- Leak channel
D- Phosphorylation-gated channel
E- Ligand-gated channel

Answer E. It opens when a chemical (ACh) binds. ***Slide:*** *Electrical communication, slide 32.*

8- Mature neurons cannot divide because they lack

A- A centrosome
B- Nissl granules
C- A Golgi apparatus
D- A nucleus
E- Mitochondria

Answer A. Neurons have a nucleus, Nissl granules, Golgi and mitochondria, but no centrosome. ***Slide:*** *Electrical communication, slide 22.*

9- Nissl granules are absent from the

A- Dendrites
B- Cell body
C- Axon and axon hillock
D- Proximal dendrites only
E- Perikaryon around the nucleus

Answer C. Nissl granules (rough ER) are in the cell body and dendrites. That is why the axon cannot make protein. ***Slide:*** *Electrical communication, slide 23.*

10- A patient with kidney failure has a high extracellular K^+ concentration. What happens to the resting membrane potential of his neurons?

A- It does not change

B- It becomes less negative (depolarized)

C- It reverses to about +60 mV

D- It becomes equal to the Na^+ equilibrium potential

E- It becomes more negative (hyperpolarized)

Answer B. The resting potential sits near EK. Raising outside K^+ reduces the gradient driving K^+ out, so EK and the resting potential become less negative. ***Slide:*** *Electrical communication, slides 48 and 49.*

11- A stimulus depolarizes a neuron's axon hillock from -70 mV to -62 mV. What happens?

A- The neuron hyperpolarizes

B- A full action potential fires

C- A graded potential occurs but no action potential fires

D- A small action potential fires

E- Voltage-gated Na^+ channels inactivate

Answer C. Threshold is about -50 to -55 mV. Below it the change stays a local graded potential; action potentials are all-or-none. ***Slide:*** *Electrical communication, slide 59.*

12- Myelin in the central nervous system is formed by

A- Schwann cells

B- Astrocytes

C- Ependymal cells

D- Microglia

E- Oligodendrocytes

Answer E. Oligodendrocytes in the CNS, Schwann cells in the PNS. ***Slide:*** *Electrical communication, slide 69.*

13- An action potential normally travels in only one direction along the axon because

A- The Na^+/K^+ pump is located only at the terminal

B- The axon has no Na^+ channels near the cell body

C- Myelin blocks backward current

D- Na^+ channels behind it are inactivated

E- K^+ channels ahead of it are closed

Answer D. Inactivated Na^+ channels cannot reopen straight away, so the patch behind the action potential cannot fire again. ***Slide:*** *Electrical communication, slide 66.*

14- Which event directly triggers the exocytosis of synaptic vesicles?

A- Entry of Ca^{2+} into the presynaptic terminal

B- Opening of postsynaptic Na^+ channels

C- Binding of transmitter to postsynaptic receptors

D- Entry of Cl^- into the terminal

E- Breakdown of ACh by acetylcholinesterase

Answer A. Six events: action potential, Ca^{2+} entry, release, receptor binding, ions enter, graded potential. ***Slide:*** *Chemical communication, slides 14 and 18.*

15- Acetylcholine excites skeletal muscle but slows the heart. This is because

A- ACh is broken down faster in the heart
B- ACh acts on the heart only through noradrenaline
C- The effect depends on the receptor type (nicotinic or muscarinic)
D- Different neurons release different forms of ACh
E- Cardiac muscle has no ACh receptors

Answer C. Nicotinic receptors at the NMJ are excitatory; muscarinic receptors in the heart are inhibitory. ***Slide:*** *Chemical communication, slide 30.*

16- Compared with direct (channel-linked) receptors, receptors that act through G proteins and cAMP produce responses that are

A- Independent of second messengers
B- Slower and longer-lasting
C- Always inhibitory
D- Limited to acetylcholine
E- Faster and shorter

Answer B. Indirect receptors (biogenic amines, peptides, gases) give slow synapses; direct receptors (ACh, amino acids) give fast synapses. ***Slide:*** *Chemical communication, slides 31 and 32.*

17- An unvaccinated farmer develops lockjaw and painful muscle spasms after a deep wound. Tetanus toxin causes these signs by

A- Blocking presynaptic Ca^{2+} channels at the NMJ
B- Preventing GABA release from inhibitory neurons
C- Blocking nicotinic receptors at the end plate
D- Increasing glutamate reuptake
E- Destroying acetylcholinesterase

Answer B. Loss of inhibition leaves excitation unopposed. ***Slide:*** *Chemical communication, slide 38.*

18- Habituation to a repeated harmless stimulus results mainly from

A- Faster conduction in the afferent axon
B- Release of serotonin onto the sensory terminal
C- Reduced activity of presynaptic voltage-gated Ca^{2+} channels
D- Increased number of postsynaptic receptors
E- Blocking of K^{+} channels by a cAMP cascade

Answer C. Habituation depresses synaptic activity. The K^{+} channel block through serotonin and cAMP is the mechanism of sensitization. ***Slide:*** *Chemical communication, slide 63.*

19- Two-point discrimination is much better on the fingertip than on the lower leg because the fingertip has

A- Fewer sensory receptors
B- Only nociceptors
C- Larger receptive fields
D- Small, densely packed receptive fields
E- Slower afferent fibres

Answer D. Small, dense receptive fields plus lateral inhibition sharpen localisation; the cortical area matches the sensitivity. ***Slide:*** *Chemical communication, slides 71 and 72.*

20- Compared with chemical synapses, electrical synapses

A- Can amplify and be modulated by drugs
B- Are always one-way
C- Use neurotransmitter vesicles
D- Pass current through gap junctions with almost no delay
E- Have the longest synaptic delay

Answer D. Electrical synapses are fast and can conduct both ways. Chemical synapses are one-way, have a synaptic delay and can be modulated.

21- Myosin II is a hexamer made of

A- Two heavy chains and two light chains
B- Three heavy chains and three light chains
C- Four heavy chains and two light chains
D- Two heavy chains and four light chains
E- Six identical light chains

Answer D. The head (S1) holds the ATP and actin binding sites; the neck is the lever arm. ***Slide:*** *Muscle physiology I, slide 14.*

22- The dihydropyridine (DHP) receptor of skeletal muscle is

A- A voltage-gated Ca^{2+} channel in the T-tubule membrane
B- A Ca^{2+}-binding subunit of troponin
C- A Ca^{2+} release channel of the sarcoplasmic reticulum
D- The acetylcholine receptor of the end plate
E- The Ca^{2+} pump that refills the sarcoplasmic reticulum

Answer A. The DHP receptor gates the ryanodine receptor (SR release channel); SERCA pumps Ca^{2+} back. ***Slide:*** *Muscle physiology I, slide 27.*

23- Which statement about the sliding filament mechanism is correct?

A- The thin filaments shorten
B- The thick filaments shorten
C- The H zone narrows while the A band stays the same
D- The A band narrows
E- The I band widens

Answer C. Filaments keep their length and slide; the I band and H zone shorten. ***Slide:*** *Muscle physiology I, slide 39.*

24- After six months of strength training, a weightlifter's biceps is much larger. This is mainly due to

A- An increase in the number of motor neurons
B- Conversion of smooth muscle into skeletal muscle
C- An increase in the number of fibres (hyperplasia)
D- Fusion of new myoblasts into extra fibres
E- Enlargement of existing fibres (hypertrophy)

Answer E. The number of skeletal muscle fibres is fixed at birth. ***Slide:*** *Muscle physiology I, slide 20.*

25- Which protein spans half a sarcomere, anchors thick filaments to the Z disc and gives the sarcomere elasticity?

A- Tropomyosin
B- Troponin
C- Nebulin
D- Actin
E- Titin

Answer E. Titin is structural and elastic; nebulin is non-elastic along actin; tropomyosin and troponin are regulatory. ***Slide:*** *Muscle physiology I, slide 27.*